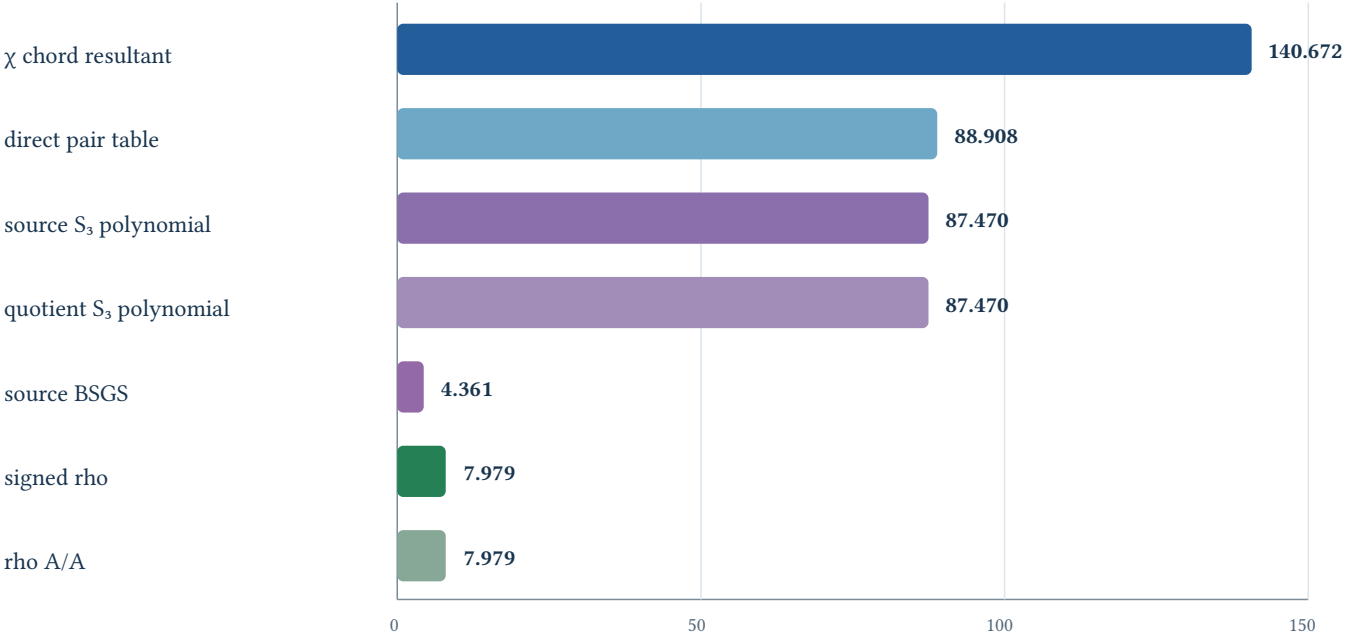
Held-out order-229 solves: seven native arms

Four targets × three rounds each · mean cold S_group = additions plus doublings / √229



All four relation arms: identical trial and relation support, rank nine, and verified scalars in 12/12 paired cells.

Source: HELDOUT_SEVEN_ARM_V3.summary.json; 84 raw children and independent checker receipt. Portable S unset.